

EXPERIMENT - 06

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Semester: 5	Date of Performance: 10/08/2026
Subject Name: ADBMS	Subject Code: 24CSP-310

(A) Problem Statement

A fruit distribution company records the number of apples and oranges sold each day. For every date, exactly one record exists for apples and one record exists for oranges.

Generate a daily sales comparison report showing the difference between the number of apples sold and the number of oranges sold on the same day. The difference should be calculated as:

Difference = Apples Sold - Oranges Sold

Display the sale date and the calculated difference. The result should be sorted by sale date in ascending order.

```
SELECT
sale_date,
SUM(
    CASE
        WHEN fruit = 'apples' THEN sold_num
        ELSE -sold_num
    END
)
AS
diff
FROM
Sales
GROUP BY
sale_date;
```

(B) Problem Statement

A company stores hierarchical employee reporting information in a table. Each record represents a node in a hierarchy where:

- `id` represents the current node.
- `p_id` represents the parent node.

Classify every node into one of the following categories:

- **Root:** A node that does not have any parent.
- **Leaf:** A node that has a parent but does not have any children.
- **Inner:** A node that has both a parent and at least one child.

Display the node ID along with its category. Return the result sorted by node ID.

```
SELECT
t1.id,
(
    CASE
        WHEN t1.p_id is NULL THEN 'Root'
        WHEN t1.id not in (
            SELECT t.p_id
            from
            Tree as t
            where t.p_id is not null
        ) THEN 'Leaf'
        ELSE 'Inner'
    END
) AS Type
FROM
Tree as t1;
```